

IQRASEC ACADEMY

WEB APPLICATION PENETRATION TESTING (WAPT/VAPT) INTERNSHIP

Task 1: Web-Application-Basics

Phase 1: Practical Web Security

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1. Objective

The objective of this task was to understand the basic working of web applications and communication between a client and a web server. So I learned the fundamentals of HTTP and HTTPS, requests and responses, HTTP methods, headers, cookies, sessions, parameters, authentication, status codes and basic APIs. So I practiced these concepts through the assigned TryHackMe rooms.

2. Concept / Understanding

HTTP and HTTPS

HTTP is a protocol used for communication between web browsers and web servers. HTTPS provides encrypted communication using TLS.

Requests and Responses

A browser sends an HTTP request to a server. The server processes the request and returns an HTTP response.

HTTP Methods

I learned common methods such as:

- GET:- It is used to request data.
- POST:- It is used to send data.
- PUT:- It is used to update data.
- DELETE:- It is used to delete data.

Headers

HTTP headers provide additional information about a request or response. They can contain information about the browser, content type, cookies, authentication, and other communication details.

Cookies and Sessions

Cookies are small pieces of data stored by the browser. They can be used to maintain information about a user. Sessions are commonly used to maintain a user's state while using a web application.

Parameters

Parameters are values sent to the web application through URLs, forms, requests or other inputs. They can influence how the application processes a request.

Authentication

Authentication is the process of verifying the identity of a user, commonly through a username and password or other authentication mechanisms.

Status Codes

HTTP status codes indicate the result of a request.

Examples include:

- 200 – OK
- 301/302 – Redirection
- 400 – Bad Request
- 401 – Unauthorized
- 403 – Forbidden
- 404 – Not Found
- 500 – Internal Server Error

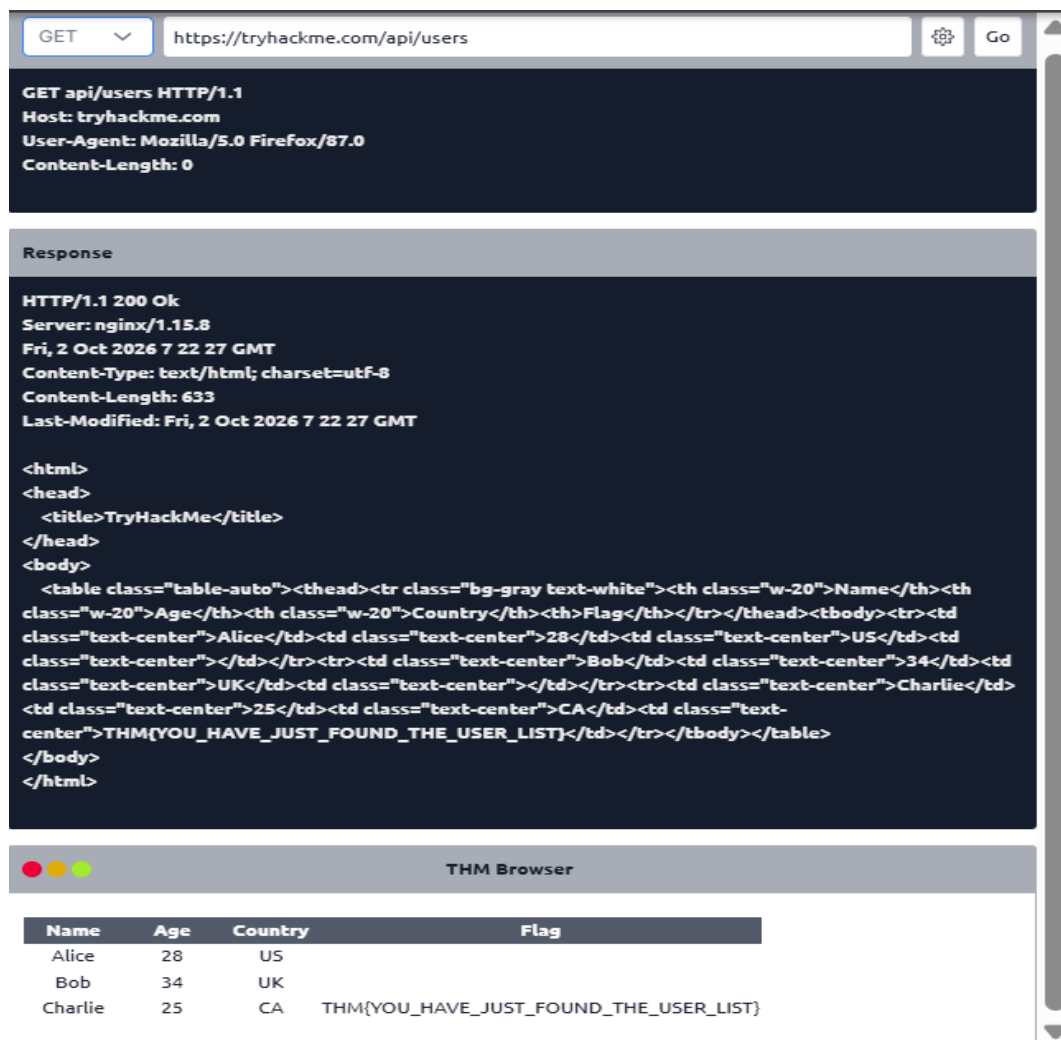
Basic APIs

I learned that APIs allow different applications or components to communicate with each other. Web APIs commonly use HTTP requests and responses to exchange data.

3. Practical Work

Room 1: Web Application Basics

I completed the TryHackMe Web Application Basics room. I studied how web applications work and learned about clients, servers, HTTP communication, requests, responses, cookies, sessions, parameters and authentication.



The screenshot shows a web browser window with the address bar set to `https://tryhackme.com/api/users`. The browser's developer tools are open, displaying the network tab. A GET request to `api/users` is shown with the following details:

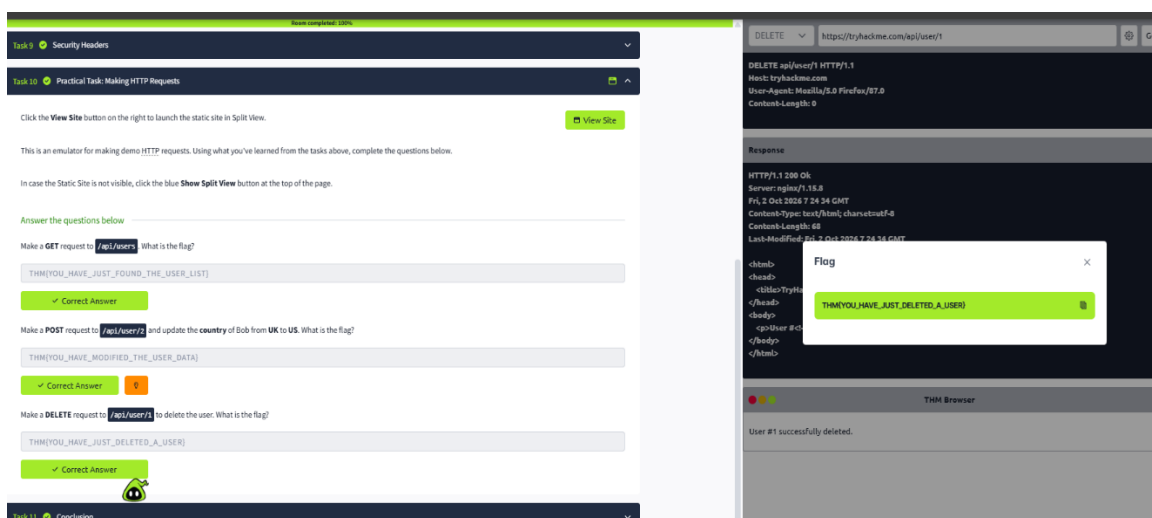
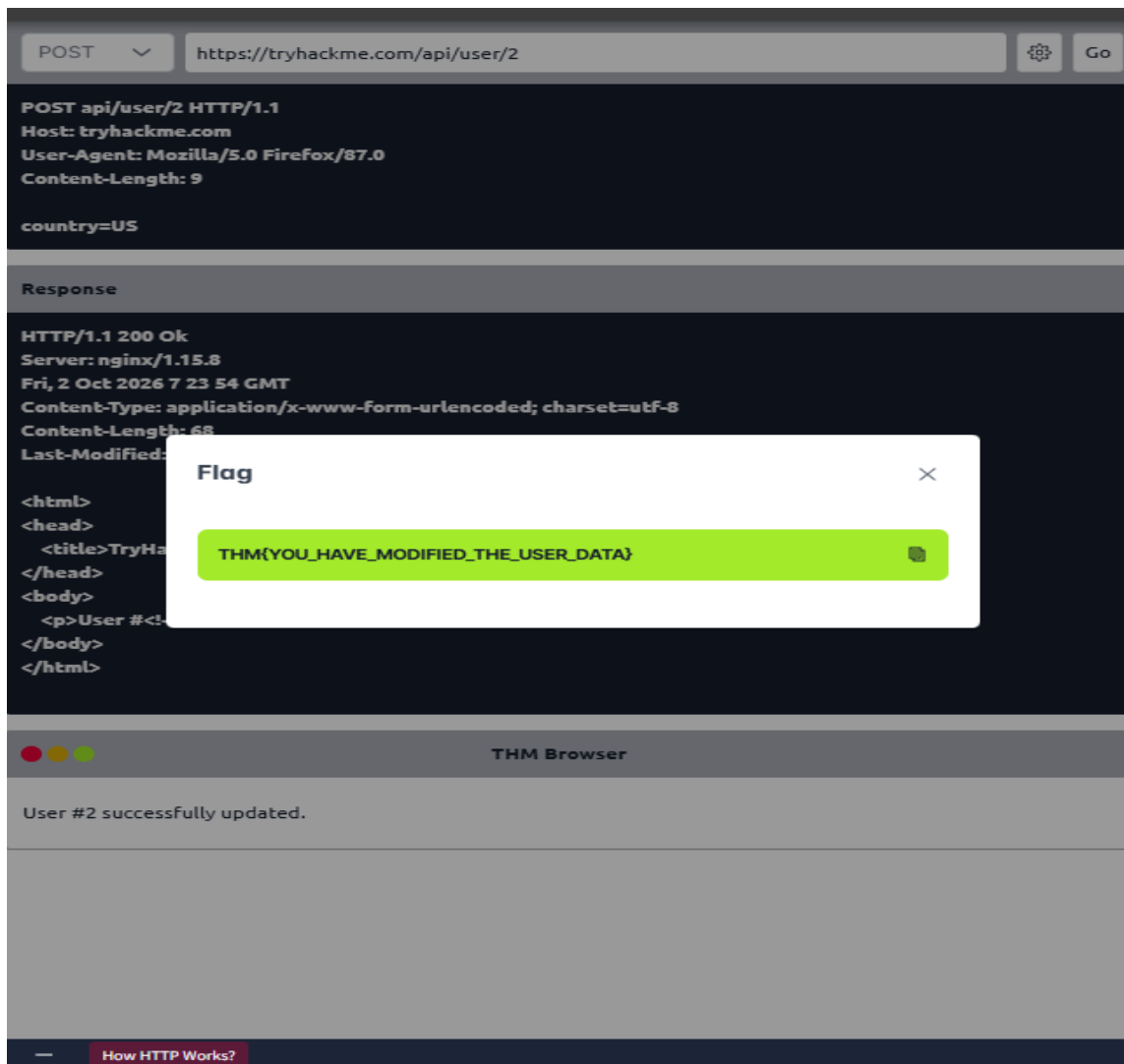
- Method: GET
- URL: `https://tryhackme.com/api/users`
- Status: 200 Ok
- Server: `nginx/1.15.8`
- Date: `Fri, 2 Oct 2026 7 22 27 GMT`
- Content-Type: `text/html; charset=utf-8`
- Content-Length: `633`
- Last-Modified: `Fri, 2 Oct 2026 7 22 27 GMT`

The response body is an HTML document with the following structure:

```
<html>
<head>
  <title>TryHackMe</title>
</head>
<body>
  <table class="table-auto">
    <thead>
      <tr class="bg-gray text-white">
        <th class="w-20">Name</th>
        <th class="w-20">Age</th>
        <th class="w-20">Country</th>
        <th>Flag</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td class="text-center">Alice</td>
        <td class="text-center">28</td>
        <td class="text-center">US</td>
        <td class="text-center"></td>
      </tr>
      <tr>
        <td class="text-center">Bob</td>
        <td class="text-center">34</td>
        <td class="text-center">UK</td>
        <td class="text-center"></td>
      </tr>
      <tr>
        <td class="text-center">Charlie</td>
        <td class="text-center">25</td>
        <td class="text-center">CA</td>
        <td class="text-center">THM{YOU_HAVE_JUST_FOUND_THE_USER_LIST}</td>
      </tr>
    </tbody>
  </table>
</body>
</html>
```

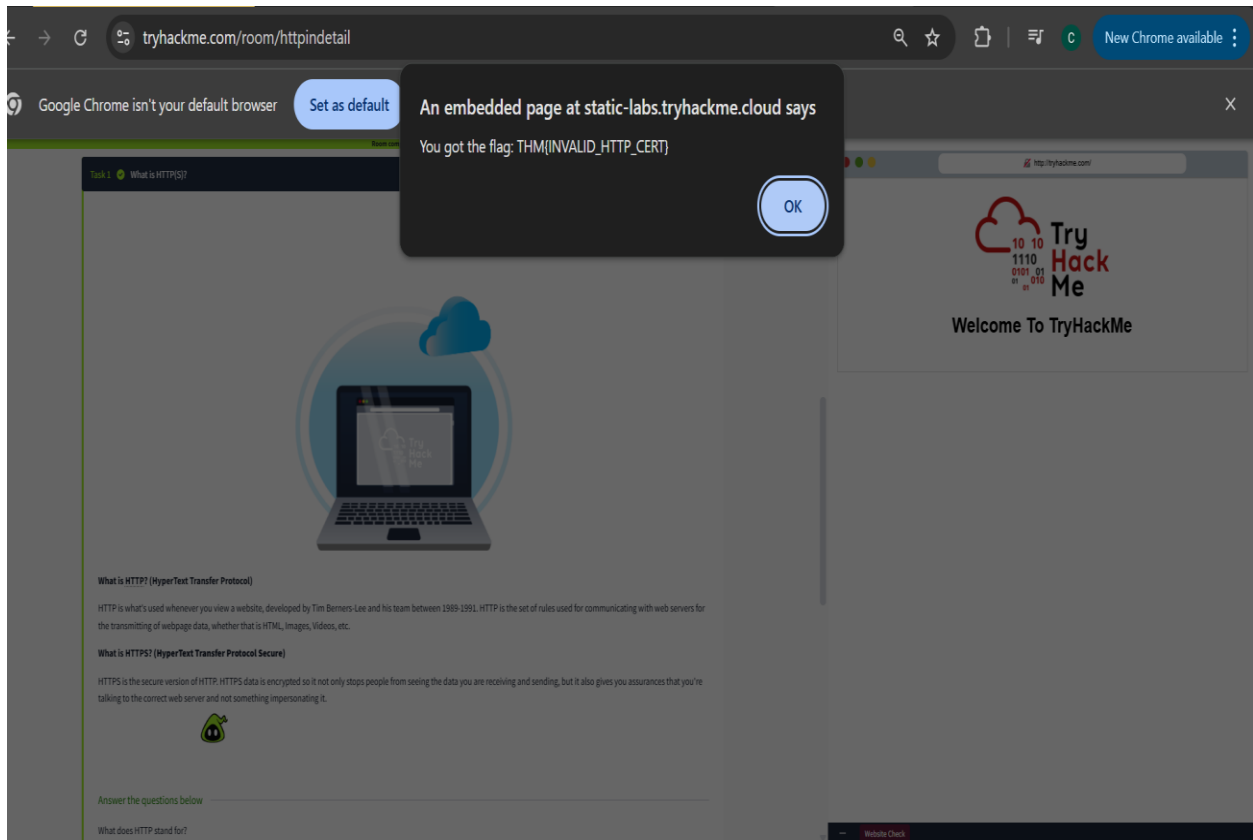
Below the browser window, a table titled "THM Browser" displays the user data extracted from the response:

Name	Age	Country	Flag
Alice	28	US	
Bob	34	UK	
Charlie	25	CA	THM{YOU_HAVE_JUST_FOUND_THE_USER_LIST}



Room 2: HTTP in Detail

I completed the TryHackMe HTTP in Detail room. I learned how HTTP requests and responses work and practiced understanding methods, headers, cookies, status codes, and other HTTP concepts.



I also reviewed the structure of HTTP requests and responses to understand how information is transferred between the browser and server.

is breakdown each line of this request:

Line 1: This request is sending the GET method (more on this in the HTTP Methods task), request the home page with / and telling the web server we are using HTTP protocol version 1.1.

Line 2: We tell the web server we want the website tryhackme.com

Line 3: We tell the web server we are using the Firefox version 87 browser

Line 4: We are telling the web server that the web page that referred us to this one is <https://tryhackme.com>

Line 5: HTTP requests always end with a blank line to inform the web server that the request has finished.

Example Response:

```
HTTP/1.1 200 OK

Server: nginx/1.15.8
Date: Fri, 09 Apr 2021 13:34:03 GMT
Content-Type: text/html
Content-Length: 98

<html>
<head>
  <title>TryHackMe</title>
</head>
<body>
  Welcome To TryHackMe.com
</body>
</html>
```

To breakdown each line of the response:

Line 1: HTTP 1.1 is the version of the HTTP protocol the server is using and then followed by the HTTP Status Code in this case "200 OK" which tells us the request has completed successfully.

Line 2: This tells us the web server software and version number.

Line 3: The current date, time and timezone of the web server.

Line 4: The Content Type header tells the client what sort of information is going to be sent, such as HTML, images, videos, pdf, XML.

Line 5: Content Length tells the client how long the response is, this way we can confirm no data is missing.

Line 6: HTTP response contains a blank line to confirm the end of the HTTP response.

Lines 7-14: The information that has been requested, in this instance the homepage.

Answer the questions below

What HTTP protocol is being used in the above example?

HTTP/1.1

✓ Correct Answer

What response header tells the browser how much data to expect?

Content Length

✓ Correct Answer

4. Tools Used

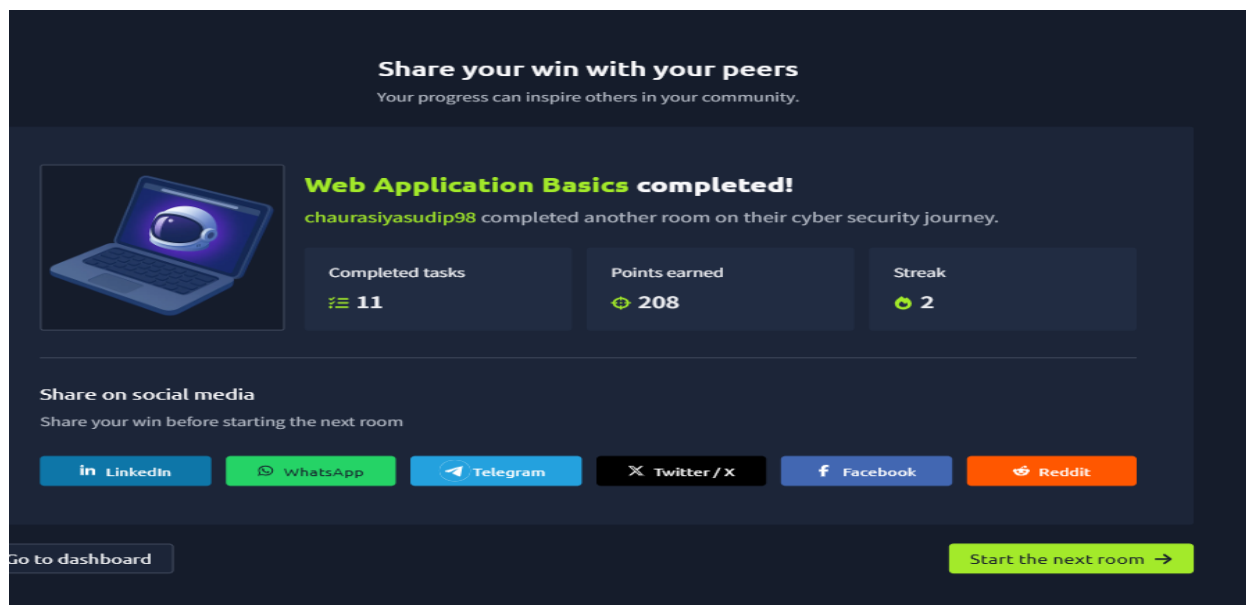
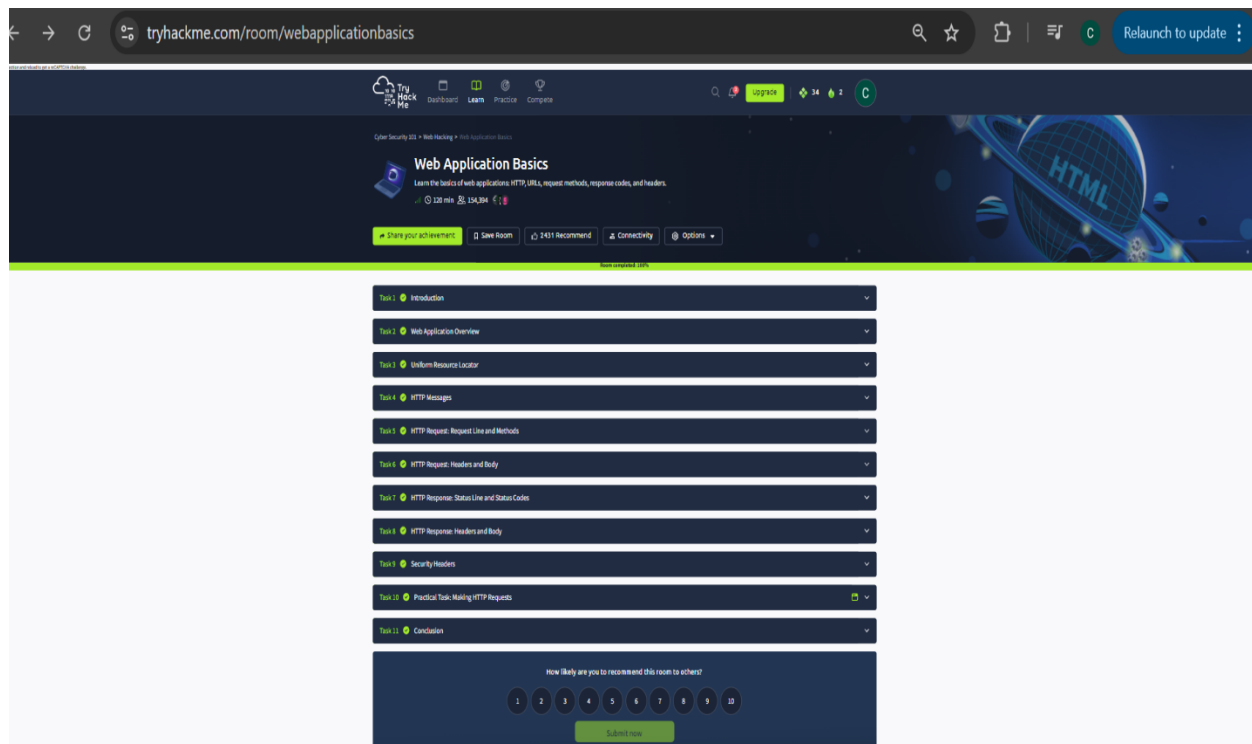
- TryHackMe
- Web Browser
- Browser Developer Tools
- HTTP request and response analysis

5. Evidence

The evidence for this task includes:

- Web Application Basics room completion screenshot.

- HTTP in Detail room completion screenshot.
- HTTP request and response practical evidence.
- Notes and observations about HTTP methods, headers, cookies, sessions, parameters, authentication and status codes.



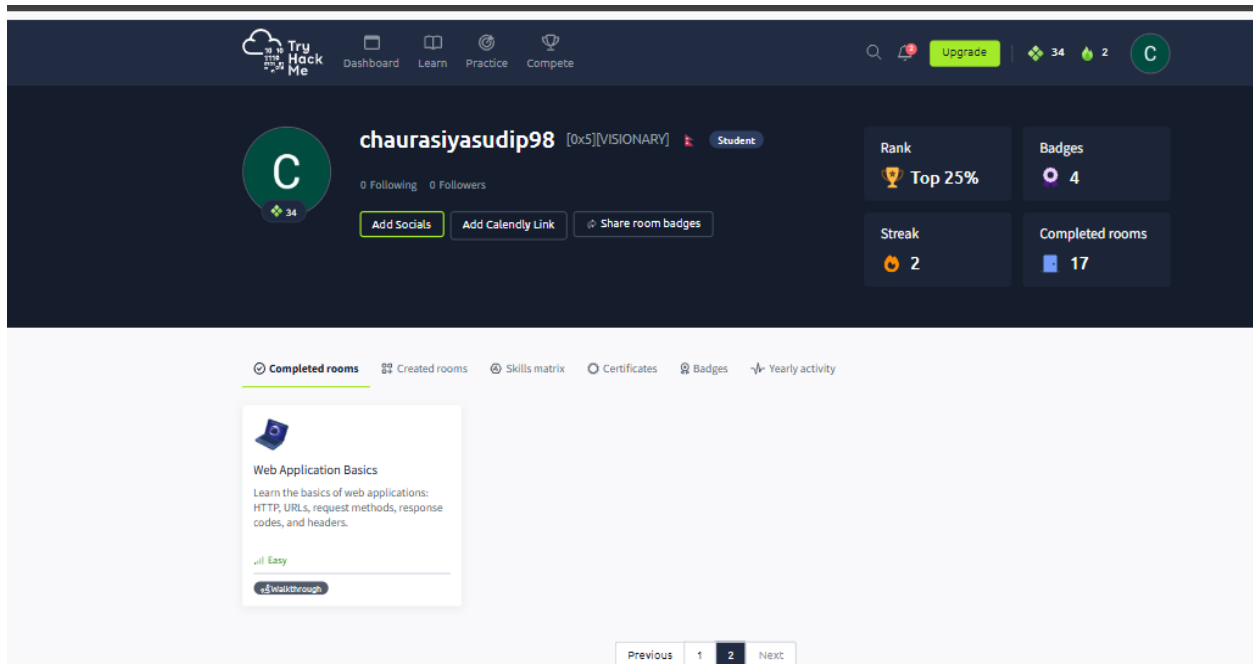
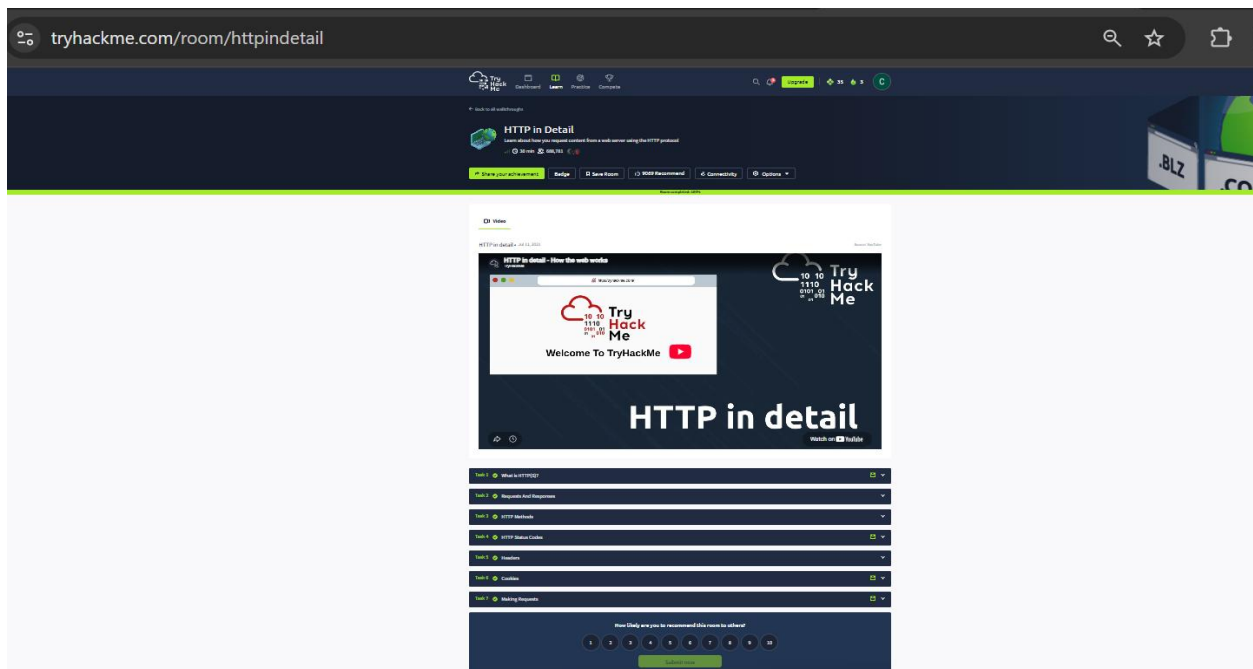


Figure: TryHackMe Web Application Basics room completed as part of Task 1.

Evidence of HTTP Completion



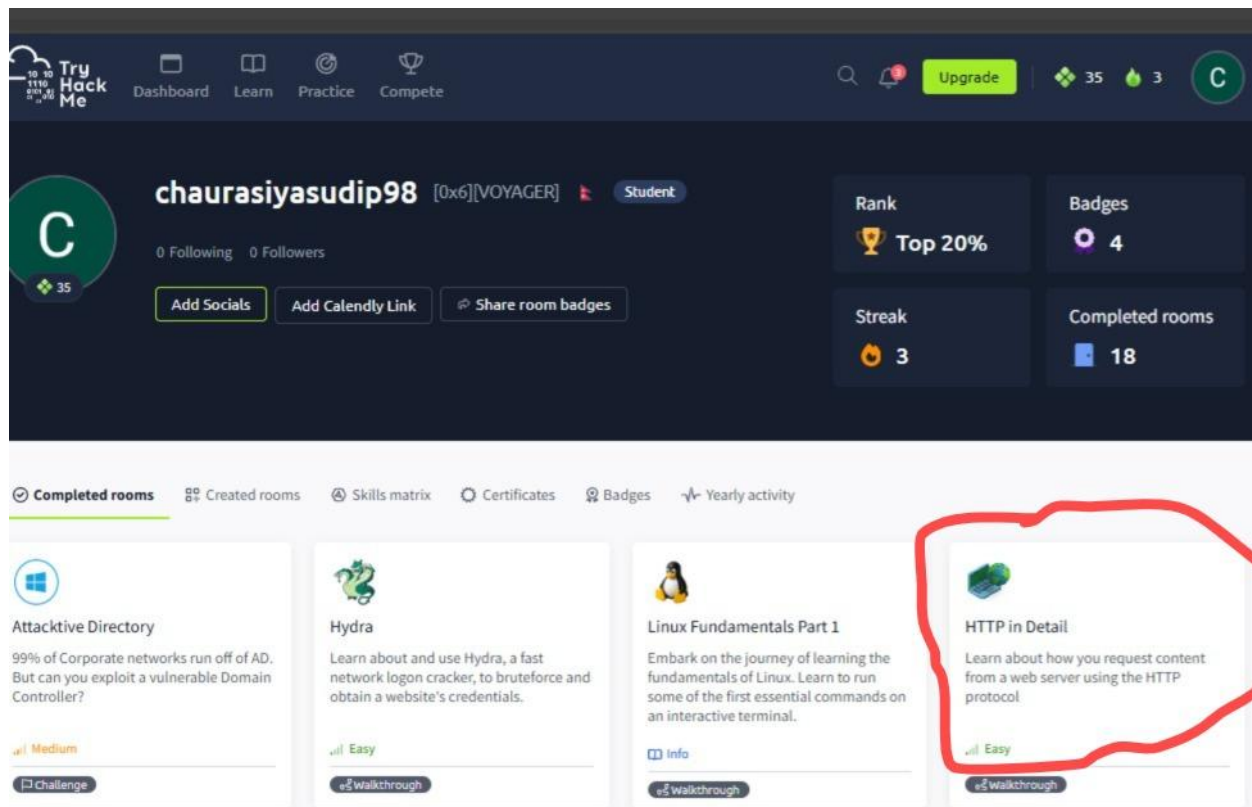


Figure: TryHackMe HTTP in Detail room completed as part of Task 1.

6. Result

I successfully completed the assigned Web Fundamentals rooms and developed a basic understanding of how web applications communicate.

I learned how HTTP requests and responses work and became familiar with important web concepts such as methods, headers, cookies, sessions, parameters, authentication, status codes, and APIs.

7. Key Learnings

- I learned how web applications communicate between clients and servers.
- I understood the difference between HTTP and HTTPS.
- I learned about HTTP requests and responses.

- I learned common HTTP methods and headers.
- I understood cookies and sessions.
- I learned how parameters are used in web requests.
- I understood the basic purpose of authentication.
- I learned common HTTP status codes.
- I learned the basic purpose of web APIs.
- These concepts will help me understand web application security testing.

TryHackMe Room completed Links

Web Application Basics:

<https://tryhackme.com/p/chaursiyasudip98>

HTTP in Detail:

<https://tryhackme.com/room/httpindetail>